

Q4. Harms of Classification (based on slide: Avoiding Harms in Classification)

Tasks:

1. Define **representational harm** and explain how the Kiritchenko & Mohammad (2018) study demonstrates this type of harm.

Representational harm occurs when a system reinforces, reflects, or amplifies harmful stereotypes or negative associations about certain groups of people. It doesn't necessarily change access to resources (that would be **allocational harm**), but it affects how groups are portrayed, perceived, and valued in society.

For example, if a sentiment analysis system consistently associates words about women, certain ethnicities, or marginalized groups with negative emotions, it perpetuates harmful stereotypes.

Kiritchenko & Mohammad (2018) Study

In their 2018 study on sentiment analysis systems, **Kiritchenko & Mohammad** demonstrated representational harm by showing that widely used sentiment and emotion lexicons systematically reflected biases:

- Phrases describing **African Americans** were scored as more negative than equivalent phrases describing **European Americans**.
- **Female identity terms** (e.g., “woman,” “girl”) were more often associated with **emotions like anger and fear**, whereas **male terms** were linked more to dominance or power.

This reveals that automated systems, when trained on biased data or lexicons, **reproduce stereotypes** — portraying certain demographic groups in disproportionately negative ways.

2. What is one risk of censorship in toxicity classification systems (based on Dixon et al. 2018, Oliva et al. 2021)?

One key risk of censorship in toxicity classification systems, as shown in **Dixon et al. (2018)** and **Oliva et al. (2021)**, is that they often **over-flag comments that simply mention marginalized identities** (e.g., “I am gay,” “Being Muslim is important to me”) as “toxic.”

This happens because the models learn spurious correlations between identity terms and toxicity from biased training data. The consequence is a form of **silencing or erasure**: legitimate self-expression, advocacy, or community discussion by marginalized groups gets suppressed, while dominant groups' speech is less likely to be misclassified.

In short: **the risk is that toxicity classifiers reinforce inequality by censoring marginalized voices more heavily than others.**

3. Give one reason why classifiers may perform worse on African American English or Indian English, even though they are varieties of English.

One key reason is mismatch between training data and linguistic variety.

Most classifiers are trained on **Standard American or British English corpora**, so they don't capture the **lexical, grammatical, and stylistic features** of African American English (AAE) or Indian English. As a result:

- Words, phrases, or syntax common in these varieties may be misinterpreted as *out-of-vocabulary*, "incorrect," or even "toxic."
- Dialect-specific meanings (e.g., "bad" meaning "good" in AAE) can be misclassified because the model has not seen enough examples in training.

In short: **classifiers perform worse because they lack exposure to the linguistic patterns of these varieties, leading to biased errors.**